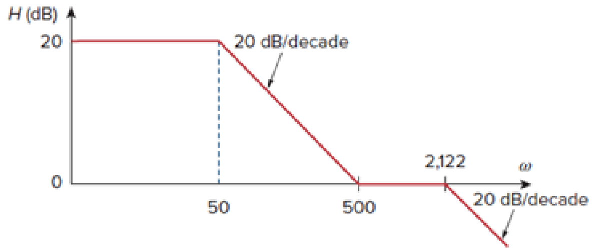
Chapter 14, Problem 24P

Problem

The magnitude plot in Fig. 14.76 represents the transfer function of a preamplifier. Find *H*(*s*).

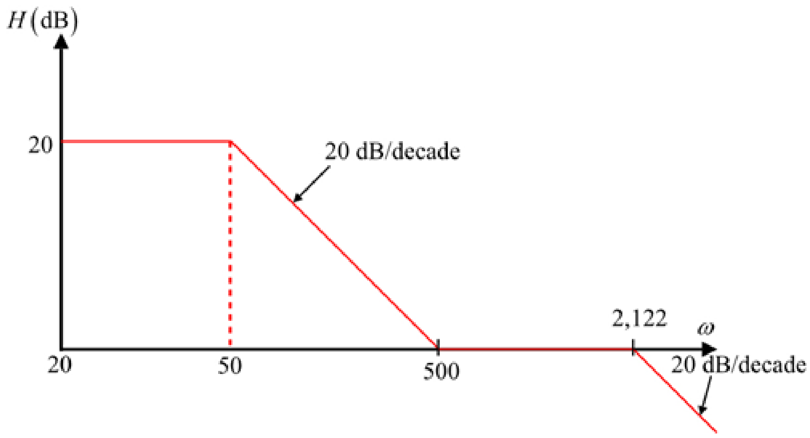
Figure 14.76:



Step-by-step solution

Step 1 of 5

Consider the following magnitude plot of the preamplifier:



Step 2 of 5

Figure 1

Step 3 of 5

To obtain $\mathbf{H}(\omega)$ from the bode plot; a zero always causes an upward turn at a corner frequency, while a pole causes a downward turn. From figure 1, there is a straight line shifted from origin to 20 dB indicates that there is a 20dB gain;

Thus,

$$20 = 20\log_{10} K$$
$$\log_{10} K = \frac{20}{20}$$

Determine the value of gain, K.

$$\begin{aligned} K &= 10^{\frac{20}{20}} \\ &= 10^1 \\ &= 10 \end{aligned}$$

Step 4 of 5

In addition to the 20 dB gain, there are three factors with corner frequencies at $\omega = 50, 500$ and 2122 rad/s.

1. A pole at $p = 50$ with slope -20 dB/decade to cause a downward turn. The pole at $p = 50$ is determined as $\frac{1}{\left(1+\frac{j\omega}{50}\right)}$.

2. A zero at $p = 500$ with slope $+20 \text{ dB/decade}$ to cause a straight line and counteract the pole at 50. The zero is $\left(1+\frac{j\omega}{500}\right)$.

3. A pole at $p = 2{,}122$ with slope -20 dB/decade to cause a downward turn and contract the zero at 500 is determined as $\frac{1}{\left(1+\frac{j\omega}{2122}\right)}$.

Step 5 of 5

Substitute all these together gives the corresponding transfer function.

$$\begin{aligned} \mathbf{H}(\omega) &= \frac{10\left(1+\frac{j\omega}{500}\right)}{\left(1+\frac{j\omega}{50}\right)\left(1+\frac{j\omega}{2122}\right)} \\ &= \frac{10\left(\frac{500+j\omega}{500}\right)}{\left(\frac{50+j\omega}{50}\right)\left(\frac{2122+j\omega}{2122}\right)} \\ &= \frac{\left(\frac{10\times 50\times 2122}{500}\right)(500+j\omega)}{(50+j\omega)(2122+j\omega)} \\ &= \frac{2122(500+j\omega)}{(50+j\omega)(2122+j\omega)} \end{aligned}$$

Replace $j\omega$ with s in the above expression.

$$\mathbf{H}(s) = \frac{2122(500+s)}{(50+s)(2122+s)}$$

Therefore, the transfer function $\mathbf{H}(s)$ of the preamplifier is $\boxed{\frac{2122(500+s)}{(50+s)(2122+s)}}$.